

FORENSIC INVESTIGATION REPORT

Report No.: KFE-24-1188

Date: October 28, 2024

Client: Ardsley Financial Group

Attention: Yolanda Prieto, SVP Corporate Services

Subject: Progressive flooring failure, Corbin Tower Levels 12 through 17

Principal investigator: Ilona Fenwick, PhD, PE

1. Assignment

Ardsley Financial Group retained Kestenbaum Forensic Engineering to determine the cause of progressive adhered flooring failure across Levels 12 through 17 of its headquarters at Corbin Tower, and to estimate the scope of remediation.

2. Background

The tenant improvement was designed in 2010 and 2011 and reached substantial completion in September 2011. The floor plate is lightweight structural concrete fill, nominal 3-1/4 inches, over 3 inch composite metal deck. Adhered carpet tile, resilient tile and resilient sheet flooring were installed on Levels 12 through 19.

Ardsley reports edge lifting first observed in the fourth quarter of 2023, becoming widespread through 2024, with odor complaints beginning in the second quarter of 2024 and two indoor air quality grievances. Level 15 was taken out of service in August 2024.

Ardsley advises that a discrete failure of approximately 2,400 square feet in the Level 19 pantry and wellness room was remediated in 2013 by localized removal, application of an epoxy vapor barrier and reinstallation. That area has not failed again.

3. Field investigation

Site visits September 16, 18 and 25, 2024. Twenty-two flooring sections lifted for observation. Sixteen in-situ relative humidity probes installed per ASTM F2170 at 40 percent slab depth, read at 72 hours.

3.1 Observations at lifted sections

- **No moisture vapor barrier, membrane, coating or primer of any kind was found at the slab surface** in any of the twenty-two sections opened on Levels 12 through 17.
- Adhesive residue is soft, brown to amber, and slick to the touch at nineteen of twenty-two locations. pH at the adhesive interface measured 10.5 to 11.8.
- Backing of lifted carpet tile shows alkali staining. Resilient tile shows edge curl and seam separation.

- Dark surface growth consistent with mold was observed at four locations on Levels 14 and 15, beneath continuously covered areas. Sampling was performed by others.

3.2 In-situ relative humidity results

Level	Probes	RH range (%)
12	3	93.2 – 94.6
13	3	94.1 – 95.5
14	3	95.0 – 96.2
15	3	94.8 – 96.4
16	2	93.7 – 95.1
17	2	93.1 – 94.9

Range across all sixteen probes: 93.1 to 96.4 percent RH.

4. Analysis

The failure mechanism is **alkaline hydrolysis of the flooring adhesive**, commonly called saponification. Moisture migrating upward through the slab transports alkaline hydroxides to the adhesive layer. The adhesive is chemically degraded, loses cohesive strength, and bond is lost. The observed adhesive condition, the interfacial pH of 10.5 to 11.8, and the alkali staining on the carpet backing are all diagnostic of this mechanism.

The driving condition is the moisture state of the slab. Lightweight concrete fill over composite metal deck can dry only through its upper face. Once an adhered, relatively impermeable floor covering is installed, that drying path is closed and the slab does not dry further. In this building the slab has been at or above 93 percent relative humidity continuously since installation and will remain so indefinitely.

The industry threshold for installing adhered floor coverings without mitigation is 85 percent relative humidity by ASTM F2170. The measured condition exceeds that threshold by eight to eleven points at every location tested.

We are not aware of any condition in this building — roof leak, plumbing failure, envelope breach or slab crack — that would account for the observed moisture. The moisture is residual construction moisture in the lightweight fill, which is the expected condition for this assembly and is the condition the industry addresses with a vapor mitigation system at the time of installation.

5. Conclusions

1. The adhered flooring assemblies on Levels 12 through 17 were installed **directly over the concrete substrate with no moisture vapor mitigation of any kind.**
2. The slab was, and remains, far above the relative humidity at which adhered floor coverings can be installed without mitigation.
3. The assembly as constructed **was not capable of performing over the service life of the installation** and its failure was inevitable rather than accidental.
4. The 2013 Level 19 remediation, which included an epoxy vapor barrier, has not failed, which is consistent with conclusions 1 through 3.

This report does not address which party selected the assembly, directed any change to it, or is contractually responsible for it. Those are questions of contract and of record, not of engineering, and Kestenbaum expresses no opinion on them.

6. Remediation scope and order of magnitude

Full removal and replacement of flooring on Levels 12 through 17, approximately 106,400 square feet, including shot blasting, epoxy moisture vapor mitigation, new floor coverings, base and transitions, phased around occupancy. **Localized abatement where growth is confirmed is carried by others and is excluded from the figure below.**

Order of magnitude construction cost, excluding relocation, abatement performed by others, and consultant fees: **\$1,180,000.**

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